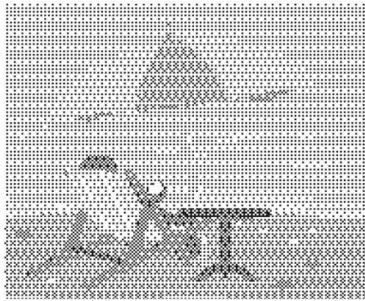


Digitization

The process of transforming information into a format that can be read by computers is called digitization. The source material can be anything, from a sound, to text, or an image, and the digitization process creates a representation of it in binary code. It does this by sampling the source material thousands of times, if not more, to come up with a series of individual binary values for each part of it.



1-bit (black-and-white)



8-bit (black-and-white)



8-bit (colour)

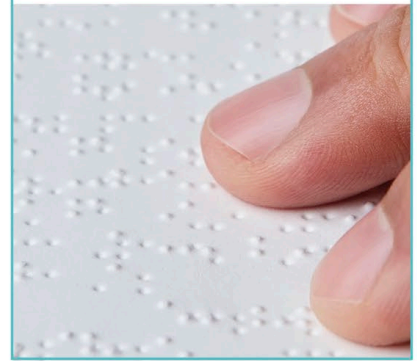


24-bit (colour)

REAL WORLD

Braille

Braille, developed by French inventor Louis Braille (1809–1852), is a famous example of binary code. It allows blind people to read by converting text into a pattern of raised dots embossed on paper. Each character is represented by a group of six dots, which can have the binary values of “raised” or “not raised”.



< Bit depth

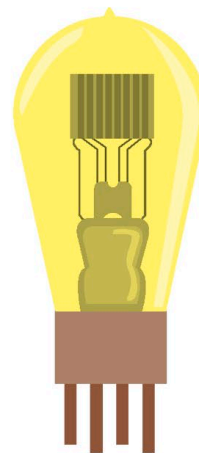
If a binary number representing a digitized value has more bits, the representation will be closer to the original. Two bits allow only four colours in a digital image, while eight bits allow 256 colours.

Binary and switches

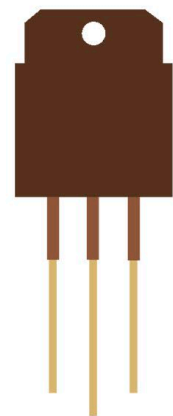
Computers have always relied on switches to achieve the flow of electric current (representing 1) or the absence of current (representing 0). The switches in the earliest electric computers were thermionic valves, which looked and behaved like light bulbs. They could also switch between allowing an electric current to flow through them and stopping it. Nowadays, the heart of a computer is a microchip, or integrated circuit, which is created using transistors.

▷ Valves and transistors

Computers made with valves took up a lot of space, often a whole room. Valves were later replaced by tiny transistors made of materials such as silicon and germanium, which also allow current to be switched on and off.



Thermionic valve



Transistor